Why randomization selection of units matters

Presentation Outline

1. **Examples: random choice of units**
   - NHANES survey
   - 1936 Literary Digest survey
   - Rock weight

Four types of studies:

Reminder:

	Treatment assignment	
Choice of units	Random	Not
Random	experiment (rare)	survey
Not	experimental study	observational study

Do the distinctions matter? **YES!**

Choice of units:

- Random sample $\Longrightarrow$
- Not random $\Longrightarrow$

Surveys in practice

Surveys seem simple: just ask questions

Very hard to do well because hard to draw a random sample

- Sampling frame: list of all items that could be sampled
  - Want frame to enumerate the population
- Then use a probability mechanism to select elements from the frame
- Simple random sample:
  - each individual in the frame has same probability of being selected
  - individuals selected independently

Example: National Health and Nutrition Examination Survey (NHANES)

Target population: civilian non-institutionalized US population
What is needed to draw a simple random sample?

- Frame: list of all non-institutionalized civilians in US
- Selection: Probability mechanism to select individuals (the sample)
- Difficult to collect data from widely scattered individuals

Multi-stage sampling procedure

- Select counties. Probability depends on population
- Select city-block size areas. Probability depends population
- Select households from a list for each city block
- Select individuals in a household

Example: 1936 Literary Digest survey of voters

Goal: predict winner of the 1936 presidential election

Literary Digest: what went wrong?

More recent research: non-response bias[1]

- poll sent to 10 million people
- 2.38 million responded
- those who hated Roosevelt more likely to respond

Choice of frame and non-response bias remain serious concerns.

[1] Squire, P. 1988, Why the 1936 Literary Digest poll failed, *Public Opinion Quarterly* 52:125-133

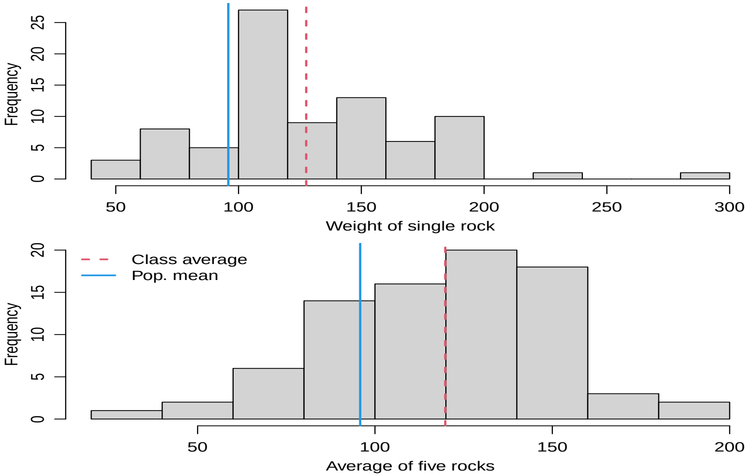
Rock weight

Context: 99 rocks, want to estimate their mean weight

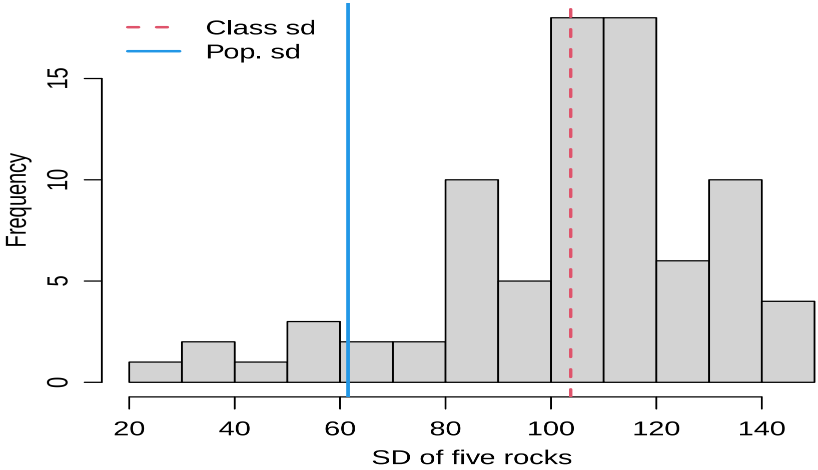
Rock weight: instructions

- Instructions to class members (when in person):
    - Select one rock that you believe best represents the mean weight
    - Select five rocks whose average best represents the mean weight
    - Select five rocks whose sd best represents the sd of the population
- All 3 are samples from the population of 99 rocks
- Question: Is any a simple random sample?
- Question: What if you closed your eyes and blindly picked rocks?

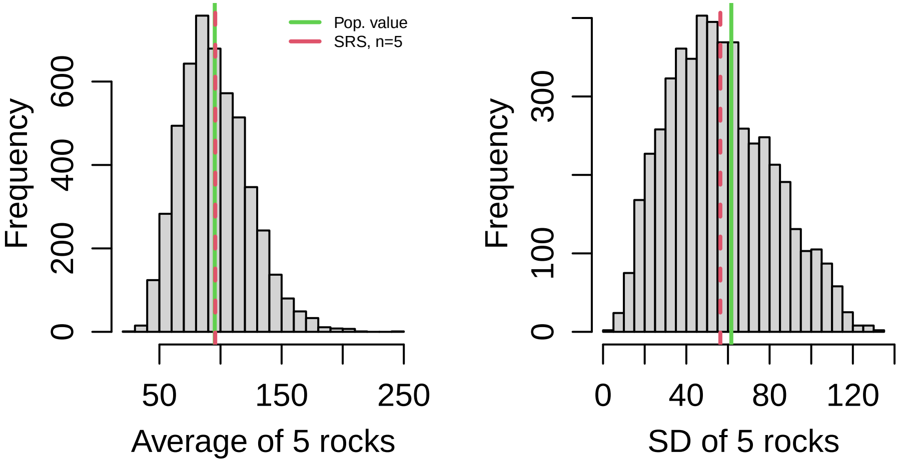
Rock weight: results for mean weight



Rock weight: results for variability



Rock weight: SRS results



Important points:

Why randomization matters

- Random choice of units:
    - Generalize from sample to population based on just the data and study design
    - Very hard to do appropriately